The Effect of Context on Information Processing of Food Cues

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Introduction

- Environmental cues, particularly visual cues, including advertisements and food displays, play a critical role in food selection[1]
- Increased environmental cues are changing our food choices where biological metabolism no longer entirely dictates eating habits[2]
- fMRI often utilized to assess brain activity when viewing certain food cues[3]
- Limited studies combining both context and food stimuli

Objective: To investigate the neural processes associated with food choices and the influence of obesogenic context in healthy, satiated individuals

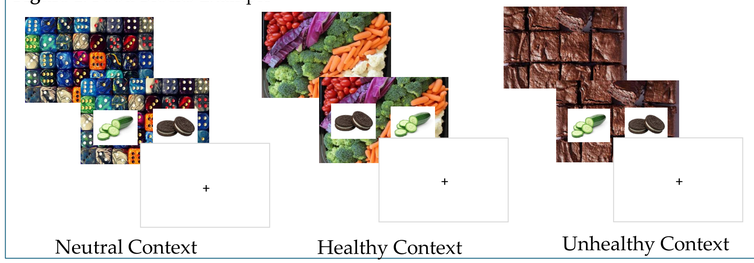
Methods

Participants:
14 healthy participants between the ages of 23-31 years old with normal BMI (18.5-24.9 kg/m^2). One female's data was excluded due to BMI criteria. Study was conducted when subjects were satiated.

Questionnaires:
- Eating Pattern Questionnaire, Power of Food Scale, TFEQ, Satiety Visual Analog Scale, Edinburgh Handedness.

fMRI Paradigm:
- Stimuli was presented through Presentation (Neurobehavioral Systems) and functional imaging data was collected from 3T Siemens scanner and statistically analyzed with SPM12
- Tasks were divided into three blocks consisting of 48 trials.
- Participants were asked to choose between two food choices, presented as pairs of one healthy and one unhealthy food within the same food type (e.g., snack, dinner)
- Each pair was presented in 3 different contexts: unhealthy, healthy and neutral backgrounds. Each pair was presented only once in a block with randomized orders of backgrounds.

Figure 1. Food Blocks Example



Results

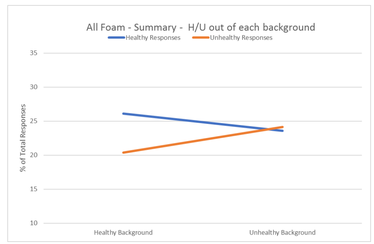


Figure 2a. Healthy vs. Unhealthy Food Responses Compared to Each Background

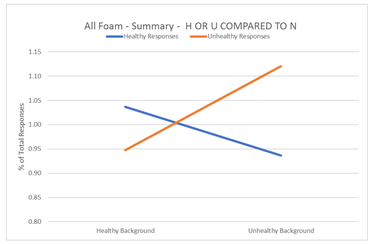


Figure 2b. Healthy vs. Unhealthy Food Responses, Controlling for Neutral Backgrounds

Figure 3a. MRI Results of Unhealthy Food Selection in Unhealthy Backgrounds

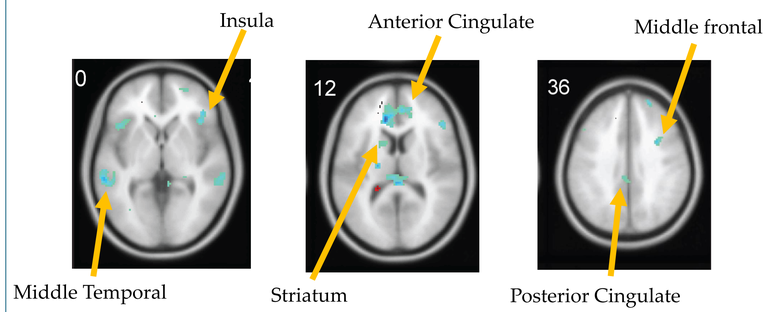
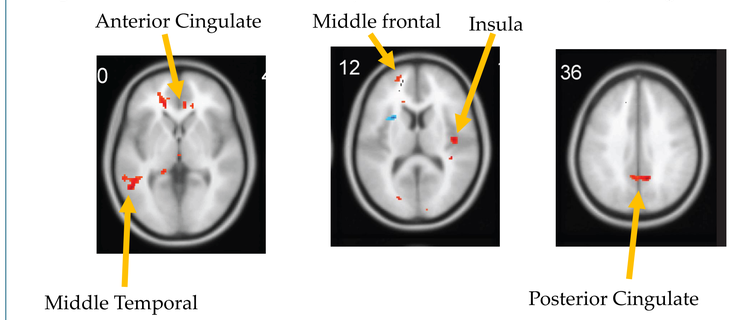


Figure 3b. MRI Results of Unhealthy Food Selection in Unhealthy Backgrounds



Results

- Behavior analysis suggests that relative to the healthy background, subjects chose more unhealthy food when seen with unhealthy backgrounds. They chose more healthy food than unhealthy food with healthy backgrounds (Figure 2a).
- Using choices from neutral backgrounds as baseline, participants were more likely to choose unhealthy food than healthy food in the healthy background. On the other hand, they chose more healthy food in healthy background (Figure 2b).
- When choosing unhealthy food in unhealthy, compared with healthy, backgrounds, activation was seen in prefrontal and cingulate cortex, which signifies stimuli selection. In addition, there was activation in the insula, striatum and temporal cortex, which is associated with food valuation (Figure 3a).
- Similarly, when choosing healthy food in healthy, compared with unhealthy, backgrounds, activation was seen in prefrontal and cingulate cortex, which signifies stimuli selection. Additionally, there was activation in the insula and temporal cortex, which is associated with food valuation (Figure 3b).

Conclusions

- Our participants were healthy young adults who, overall, made equally often healthy and unhealthy food choices. However, participants seem to be influenced by the background, and in particular, participants were more influenced by the unhealthy backgrounds than by the healthy backgrounds.
- Importantly, we observed brain activation in the striatum, which is the reward area only for unhealthy food choices in the unhealthy food backgrounds.

Limitations and Future Directions

- These results are based on relatively small number of participants, and we need to see if it will hold within a larger sample.
- We plan to continue investigating food choices with a wider range of participants, such as obese, anorexia and other conditions (e.g., hungry versus satiated)

References

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